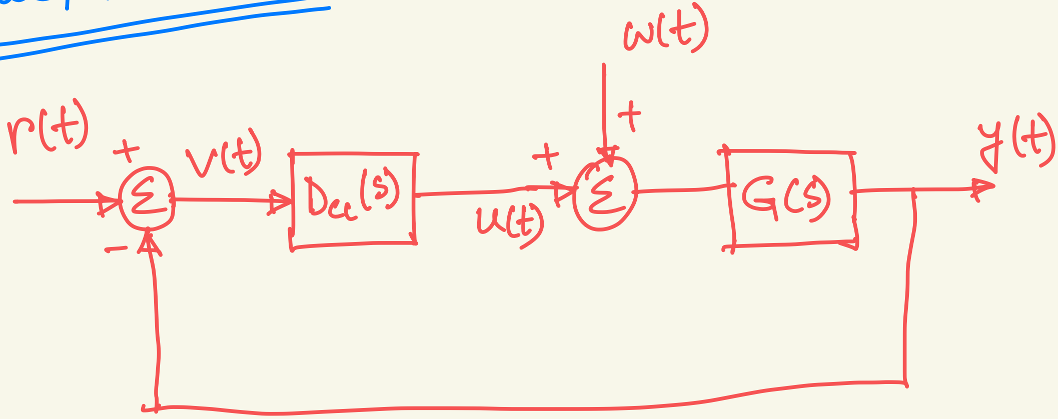
Lecture 25



From the previous lecture we know that

$$Y(s) = \frac{G(s) D_{cl}(s)}{1 + D_{cl}(s) G(s)} R(s) + \frac{G(s) \quad W(s)}{1 + G(s) D_{cl}(s)}$$

The transfer functions are rational functions.

$$G(s) = \frac{a(s)}{b(s)} \qquad D_{cl}(s) = \frac{c(s)}{d(s)}$$

$$Y(s) = \frac{a(s)\,c(s)}{b(s)\,d(s) + a(s)\,c(s)} R(s)$$

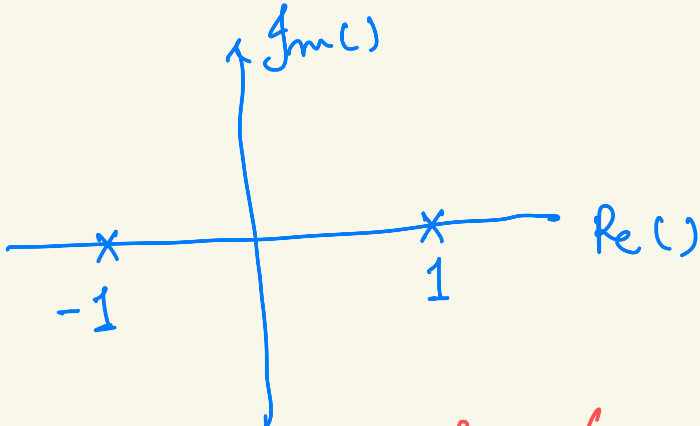
$$+ \frac{a(s)\,d(s)}{b(s)\,d(s) + a(s)\,c(s)} W(s)$$

Note that with closed loop transfer function, we have the flexibility to design $c(s)$ and $d(s)$. By proper design, we can avoid having poles in the right half plane.

Let's consider the following transfer function.

$$G(s) = \frac{1}{s^2 - 1}$$

$$= \frac{1}{(s-1)(s+1)}$$



Note that the transfer function has a pole on the right half plane. As a result, the system is unstable.

Let's form a controller that can

make the system stable.

$$D_{cl}(s) = \frac{k(s+\gamma)}{s+\delta} \qquad G = \frac{1}{s^2-1}$$

The denominator of the closed loop transfer function.

$$1 + G D_{cl} = 1 + \frac{1}{s^2-1} \cdot \frac{k(s+\gamma)}{s+\delta} = 0$$

$$\Rightarrow (s^2-1)(s+\delta) + k(s+\gamma) = 0$$

$$\gamma = 1$$

$$\Rightarrow (s-1)(s+\delta) + k = 0$$

$$\Rightarrow s^2 + (\delta-1)s + (k-\delta) = 0$$

By changing k and δ we can

have roots that will lie in the left half plane.

$$s_{1,2} = \frac{-(\delta-1) \pm \sqrt{(\delta-1)^2 - 4\cdot(k-\delta)}}{2}$$

A matlab code was shown in the class.